

SCIENCETECH COM

User Manual

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1. Introduction

1.1 Objectives

This document describes Sciencetech COM Object architecture.

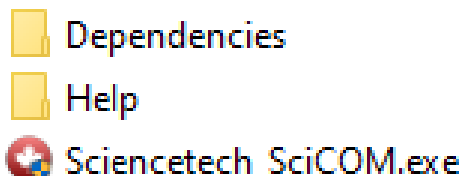
Sciencetech COM is a package that allows you to create your own application to control Sciencetech equipment. Each module will have its own set of functionalities and you will find more details about the module operation in your SciSpec user manual.

2. Installation

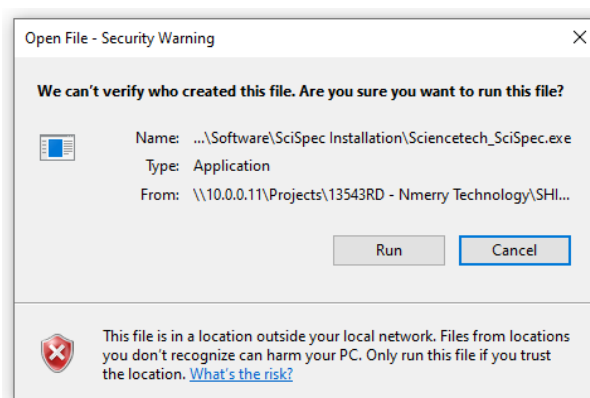
Sciencetech COM is Windows 7 and Windows 10 compatible. It may work on Windows 8 but is not thoroughly tested in that operating system.

2.1 Software Installation Sequence

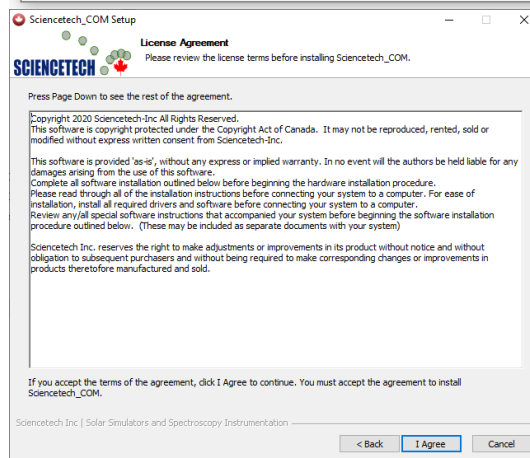
1) You can run the installer exe from a network location, USB or from the computer's hard drive. If you copy the installer to a hard drive before installation makes sure you copy the folder structure exactly. For example, in the image below you must copy all 3 items to the same directory before you can run Sciencetech_SciCOM.exe



2) Launch "Sciencetech_SciCOM.exe". In order to run, the installer will request administrator privilege. Based on your system, you will see a popup window asking for approval, please click on "Run" (Windows 7) or "Yes" (Windows 10).



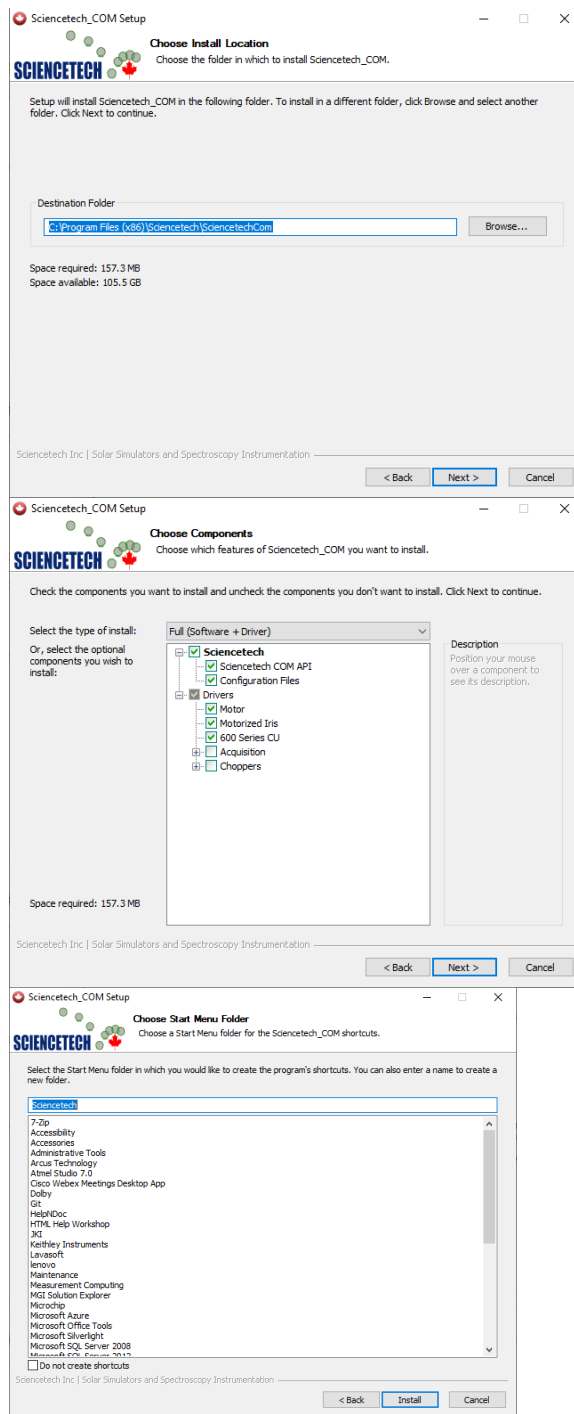
3) Review the license agreement. Click 'I Agree' to continue



4) Choose the destination folder
(By default C:\Program Files
(x86)\Sciencetech\SciencetechCom\)

5) Choose the components to be installed.
The recommended installation option is 'FULL'.
Note that some options may not be available
depending on the hardware and software that
you purchased with your Sciencetech
equipment.
Based on your decision, the installer will launch
several third-party driver installations, for each
of them when requested, **request to reboot later
and wait until Sciencetech installation had been
completed.**

6) Choose to create a StartMenu Shortcut.
This will create a shortcut for SciMono
application and SciMono User manual.
Note: At any time, a shortcut will be created on
your desktop.

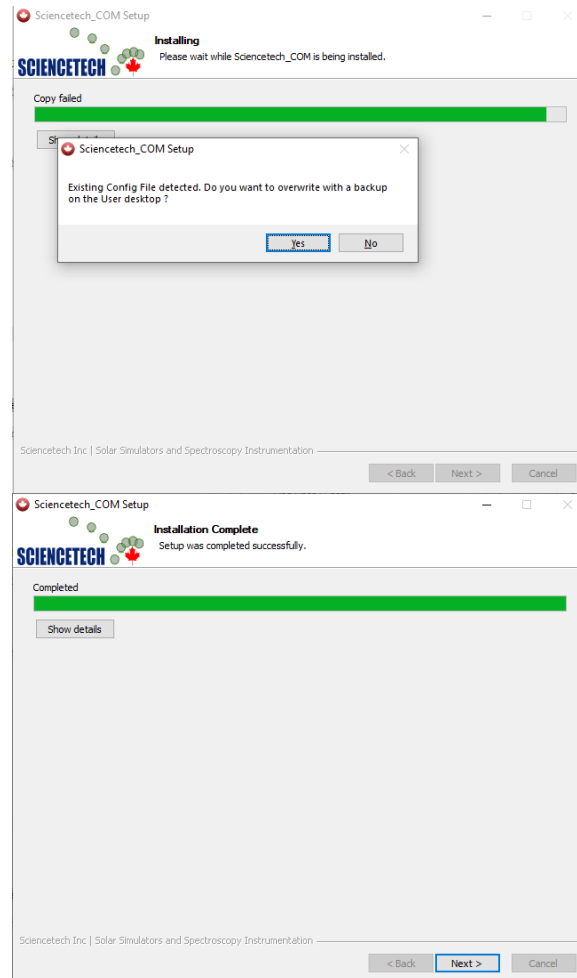


7) Continue to install the software:
If existing configuration files are detected, the installer will give you two options:

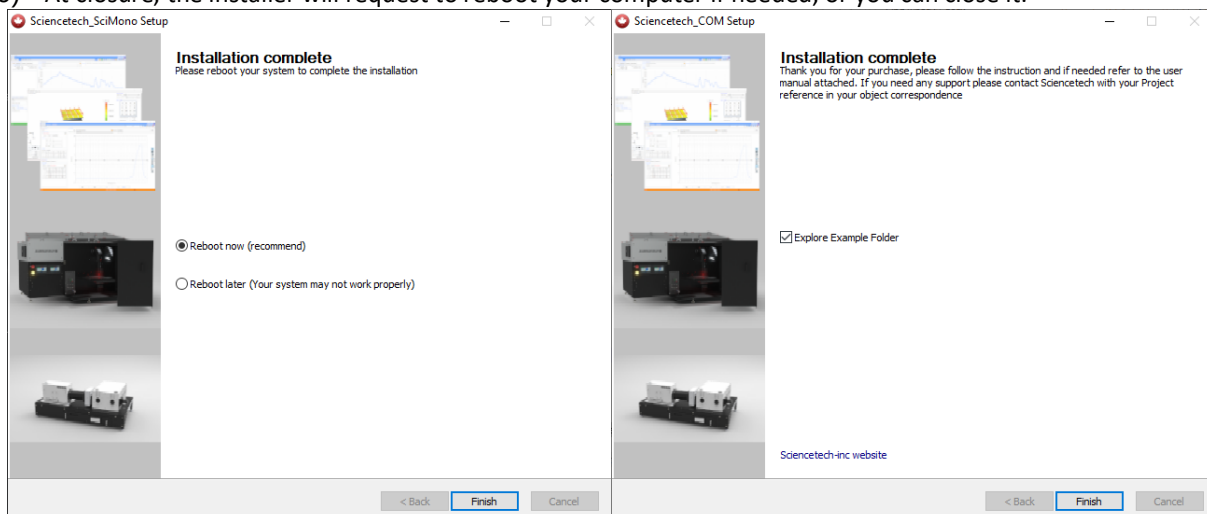
- **To not overwrite** any existing file, in that case, it is possible the old configuration file may not be compatible with this installation.
- **Overwrite the existing file** (Recommended)
The previous one will be saved in a backup folder on your desktop.

8) Successful Software Installation
When the software completed a successful installation, you will see the final popup
If the software is not successful see the troubleshooting section **Error! Reference source not found.**

3)



9) At closure, the installer will request to reboot your computer if needed, or you can close it.



10) User Manuals:

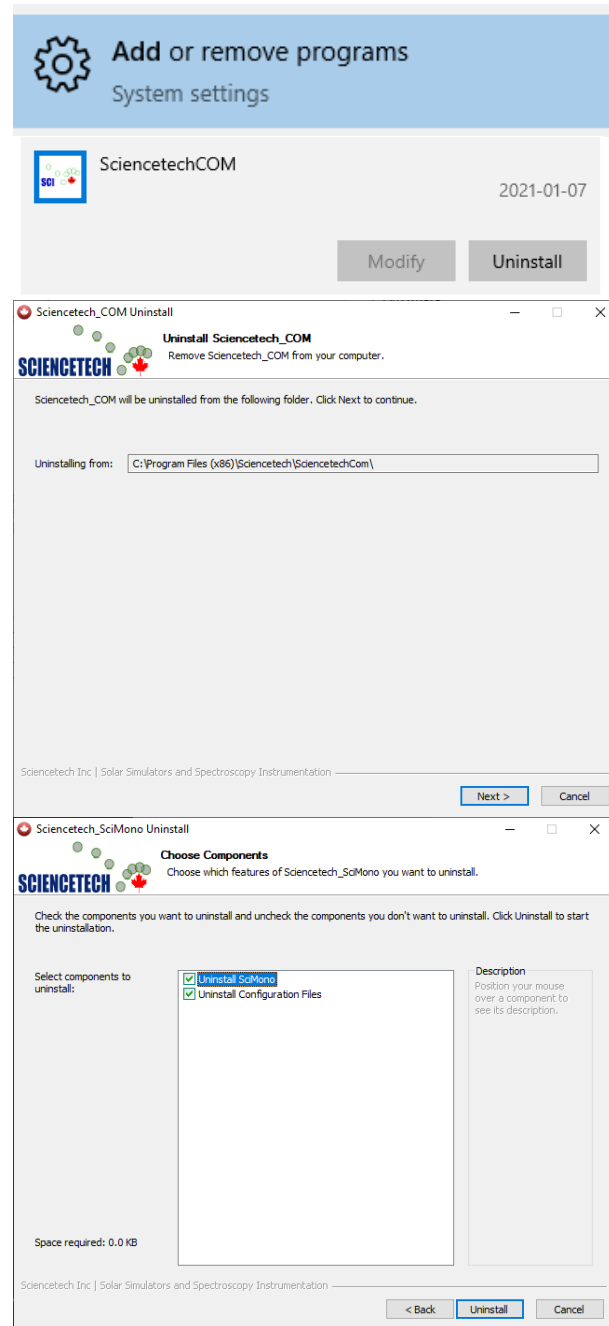
By default user manuals will be copied to the following directory: (C:\Program Files (x86)\Sciencetech\SciencetechCom\Help)

2.2 Uninstalling SciencetechCOM:

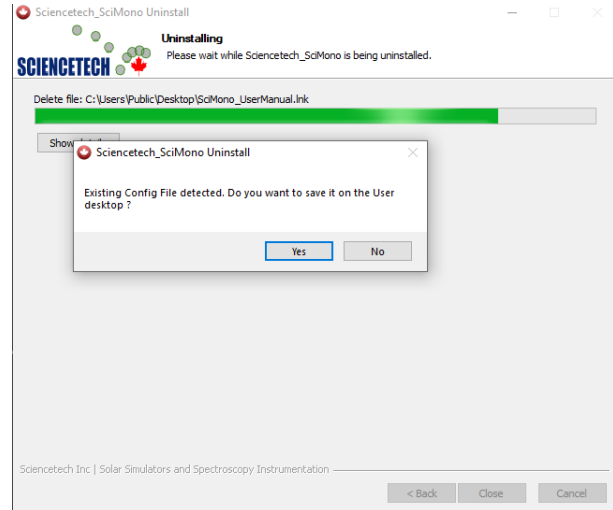
1) If you want to uninstall SciMono, please go to the Add/Remove program. Look for SciencetechCOM in the list of installed programs.

2) Click on uninstall and follow the instructions.

3) Define what you want to uninstall. Sciencetech installer will remove only Sciencetech software in order to not corrupt any third-party software. If you want to fully uninstall all the components from the installer, you will have to launch all the third-party uninstallers as well.



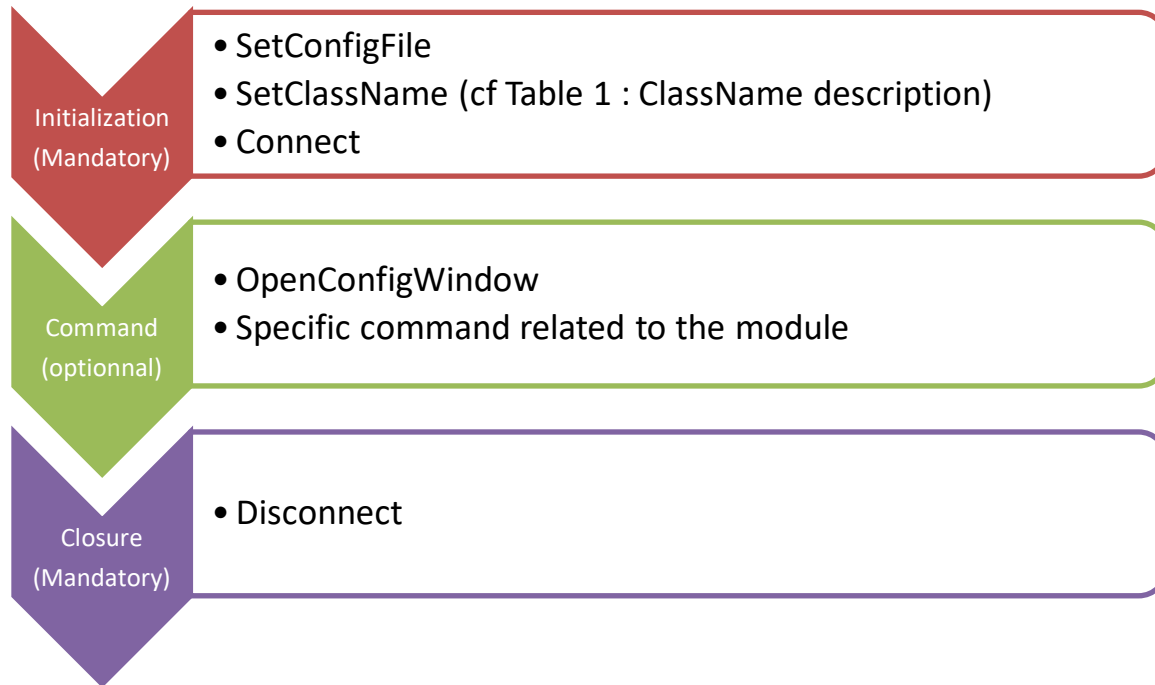
Note, if you have requested the uninstallation of the config file, a popup will ask you for confirmation and if you want to keep a backup on the desktop. (Recommended if you have modified the initial config file provided with the installation).



3. API Module description

3.1 Fundamental Sequence

In order to control your equipment, any application should match the following structure and sequence order:



3.2 Global Structure

Each module has at least the following capabilities:

3.2.1 Event

Name	Parameter	Description
OnError	String errString	Error based on each module
OnPropertyChanged	String PropName	Property change based on an event name

3.2.2 Properties

All the properties are read only. Some parameters could be configured with some methods.

Name	Type	Description
Name	String	Name of the module
IsConnected	Bool	Connection status
ErrorMessage	String	Provide the last error recorded, if no error seen, it will say "No Error"

3.2.3 Methods

All the method should return at least a Boolean "success" to acknowledge that the command had been received (asynchronous) or received and done (synchronous)

Name	Input	Output	Description
SetClassName	String ClassName String DllName	Bool Success	Set the ClassName and DLL full file path. Call the method prior to SetConfigFile to load the module.
SetConfigFile	String file Path	Bool Success	Set the configuration file and return True if successful
Connect	/	Bool Success	Connect to the module and provide an OK/KO status
OpenConfigWindow	/	Bool Success	Open the configuration window
Disconnect	/	Bool Success	Close the connection with the module and provide OK/KO status

3.3 SCIDetector_API

3.3.1 Event

Name	Parameter	Description

3.3.2 Properties

All the properties are read only. Some parameters could be configured with some methods.

Name	Type	Description
------	------	-------------

isAcquiring	Bool	Is currently acquiring signal
Frequency	Integer	Acquisition frequency (Sample per second)
SelectedChannel	Integer	Selected channel
Minimum Range	Double	Minimum signal level (unit defined by the hardware)
Maximum Range	Double	Maximum Signal level (unit defined by the hardware)

3.3.3 Methods

Name	Input	Output	Description
SetChannel	Int channel index	Bool Success	Set the active channel
SetRange	Double Min Double Max	Bool Success	Set the range signal (unit defined by the hardware)
SetFrequency	Int Frequency	Bool Success	Set the acquisition frequency (Hz)
GetRawSignal		Double Raw Signal Bool Success	Get the average value of the data BufferSize based on the IntegrationTimeMX (config file)

3.4 SCICHopper_API

3.4.1 Event

Name	Parameter	Description

3.4.2 Properties

Name	Type	Description
IsStoppedModeSupported	Bool	True if chopper can be stopped in either the open or closed position
IsMoving	Bool	Chopper moving
IsInOpenPosition	Bool	Chopper in Open fixed mode
IsInClosedPosition	Bool	Chopper in Closed Fixed mode
TriggerMode	Int	Trigger mode depends on the chopper configuration. 0 : Internal 1 : External 2 : Special
MinimumFrequency	Double	Minimum frequency available in Hz
MaximumFrequency	Double	Maximum frequency available in Hz

3.4.3 Methods

Name	Input	Output	Description
GetFrequency		Double Frequency Bool Success	Current Frequency (Hz)
SetFrequency	Double Frequency	Bool Success	Set the current chopper frequency in Hz
SetStoppedMode	Bool positionIsOpen	Bool Success	True : Chopper fixed in Open position False : Chopper fixed in Closed position (Will raise an error if IsStopped mode not supported)
SetTriggerMode	Int TriggerMode	Bool Success	Set the Trigger mode 0 : Internal / 1 : External / 2 : Special

3.5 SCILAMPPOWERSUPPLY_API

3.5.1 Event

Name	Parameter	Description

3.5.2 Properties

Name	Type	Description
IsLampOn	Bool	The lamp is currently ON
IsFanOn	Bool	The Lamp fan is ON
Current	Double	Current in Amps
Voltage	Double	Voltage in Volts
Power	Double	Power in Watts
Stabilization supported	Bool	Lamp signal stabilization available
Stabilization period	Double	Stabilization period of the lamp in second
IsStabilized	Bool	The lamp is stabilized if this feature is available
HasShutter	Bool	True if shutter available
IsShutterOpen	Bool	True = Shutter open
CurrentVoltageSettingSupported	Bool	The current and voltage could be setup
MaximumCurrent	Double	Maximum Lamp current in Amps
MaximumVoltage	Double	Maximum LampVoltage in Volts (Available only if VoltageSettingsSu
OutputIntensityLevelSupported	Bool	Output intensity level could be set directly
MinimumOutputPercentage	Double	Minimum intensity Percentage
MaximumOutputPercentage	Double	Maximum intensity percentage
PowerSettingSupported	Bool	Power level could be set directly
SunLevelSettingSupported	Bool	Sun level could be set directly

3.5.3 Methods

Name	Input	Output	Description
SetFanState	Bool State	Bool Success	True = Turn on the lamp fan False = Turn off the lamp fan
PowerLamp	Bool State	Bool Success	On= Lamp ON Off = LampOff
SetCurrentVoltage	Double Current Double Voltage	Bool Success	Set the current and voltage of the lamp If the feature is available
SetOutputPercentage	Double percentage	Bool Success	Set the output percentage If the feature is available
SetSunLevel	Double SunLevel	Bool Success	Set SunLevel If the feature is available

Stabilize	Bool State	Bool Success	True = Wait for lamp stabilization False = No dwell time If the feature is available
SetPower	Double Power	Bool Success	Set the power level If the feature is available

3.6 SciFilterwheel_API

3.6.1 Event

Name	Parameter	Description

3.6.2 Properties

Name	Type	Description
CurrentFilter	Int	Current filter index
IsHoming	Bool	FilterWheel Is currently homing
IsHomed	Bool	Filter wheel successfully homed
IsMoving	Bool	Filter wheel under motion
NumberOfFilters	Int	Number of filters on the wheel

3.6.3 Methods

Name	Input	Output	Description
Stop		Bool Success	Stop the motor motion
Home		Bool Success	Start homing the filter
MoveToFilterNumber	Int Filter Number	Bool Success	Move to the selected filter index
MoveToFilterWavelength	Double Wavelength	Bool Success	Move to the filter appropriate for the wavelength
GetFilterDescription	Int Filter Number	String Description Bool Success	String description for the filter
GetMinimumWavelength	Int Filter number	Double Min Wav. Bool Success	Minimum wavelength for the filter
GetMaximumWavelength	Int Filter number	Double Max Wav. Bool Success	Maximum wavelength for the filter

3.7 SCIMONO_API

3.7.1 Event

Name	Parameter	Description

3.7.2 Properties

Name	Type	Description
AutoGrating	Bool	True = Auto Grating False = Manual Grating
CurrentGrating	Int	Number of gratings available in the Monochromator
IsHoming	Bool	The monochromator is Homing
IsHomed	Bool	The monochromator is Homed
IsMoving	Bool	Motor in motion

3.7.3 Methods

Name	Input	Output	Description
------	-------	--------	-------------

GetDispersion	Int Grating	Double Dispersion Bool Success	Return Dispersion of the given Grating in nm (Calculation only good at 0nm)
GetWavelength		Double Wavelength Bool Success	Return the current wavelength for the monochromator
Home		Bool Success	Home the Monochromator
SetAutoGrating	Bool Auto	Bool Success	Set if the Monochromator will select the grating for the wavelength
SetGrating	Int Grating	Bool Success	Set the active grating
SetWavelength	Double Wavelength	Bool Success	Set the active wavelength (nm)
SetSpectralSpeed	Double NMPerSecond	Bool Success	Set the monochromator speed in nm/second (Used to customize the scan speed)
Stop		Bool Success	Stop any motor motion

3.8 SCISLIT_API

3.8.1 Event

Name	Parameter	Description

3.8.2 Properties

Name	Type	Description
SerialNumber	String	The serial number of the slit motor
IsHomed	Bool	Shows whether the slit is properly homed
IsMoving	Bool	Shows whether the slit is in motion
MaximumWidth	Double	Shows the maximum width the slit can be set to
MinimumWidth	Double	Shows the minimum width the slit can be set to. In most cases it will be 0
SlitWidth	Double	Shows the current width of the slit

3.8.3 Methods

Name	Input	Output	Description	Sync.
SetWidth	Double width	Bool true Bool false	Return whether the execution is successful	No

3.9 SCIIRIS_API

3.9.1 Event

Name	Parameter	Description

3.9.2 Properties

Name	Type	Description
IsHoming	Bool	The monochromator is Homing
IsHomed	Bool	The monochromator is Homed
IsMoving	Bool	Motor in motion
MinimumDiameter	Double	Minimum diameter available
MaximumDiameter	Double	Maximum diameter available
Diameter	Double	Current Iris diameter

3.9.3 Methods

Name	Input	Output	Description
Home		Bool Success	Home the Monochromator
SetDiameter	Double Diameter	Bool Success	Set the active wavelength (nm)
Stop		Bool Success	Stop any motor motion

4. Examples

4.1 Description

The “Examples” folder contains the following repositories:

- Build : Application showing a basic communication with the equipment
- Source Code : C++ project example for each module.

4.2 Interface

Each example provides a quick overview of the following steps :

- Connect to the equipment
- Catch an error
- Run a basic command

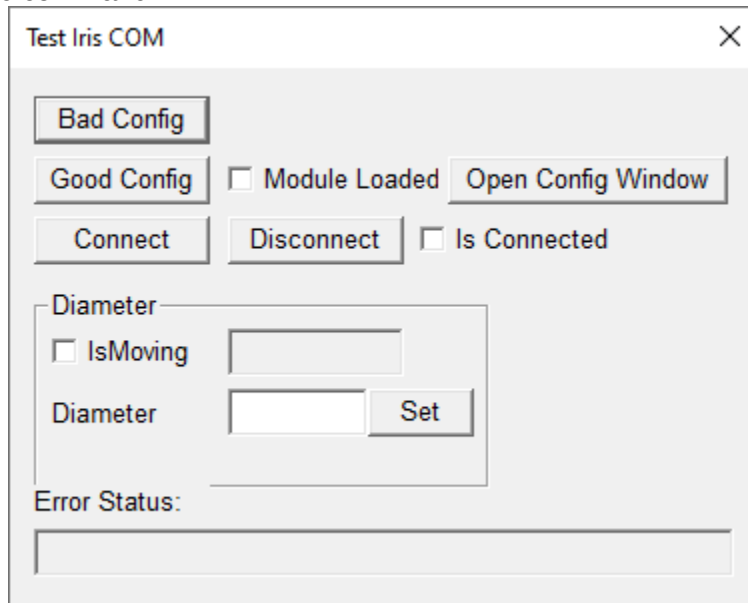


Figure 1: COM Interface Example

1. **Bad Config** : This button will load a wrong config file and show you an error message
2. **Good Config** : On click, a compliant config file we be loaded.
 - a. **If Success** : “Module Loaded” will be checked
3. **Open Config Window**: As soon as “Module Loaded” is checked, this button will open an interface giving the opportunity to the user to modify the config file.
4. **Connect**: When a correct configuration had been loaded for a specific equipment with the correct port com, it will try to connect to the equipment
 - a. **If Success** : “Is connected” will be checked
 - b. **If Error** : An error status message will be provided.
5. **Disconnect**: This will close the connection with the equipment

-
6. **Basic command** : *This part will change based on the example and the connected equipment*

4.3 Usage Notes

The following test projects are included:

- TestIrisCom: module name is SciIris
- TestChopperCom: module name is SciChopper
- TestDetectorCom: module name is SciDetector
- TestFilterWheelCom: module name is SciFilterWheel
- TestLampPowerSupplyCom: module name is SciLampPowerSupply
- TestMonoCom: module name is SciMono

Each of these projects has a dialog class namedDlg + project name. This class demonstrates the use of the COM object in C++ code. Pay particular attention to the method OnInitDialog which sets the class name and the method OnGoodConfig which attempts to load a configuration file.

For C++ development, copy the files ObjectHandler.h, ObjectHandler.cpp, <moduleName>.h, <moduleName>.cpp, <moduleName>events.h and <moduleName>events.cpp to your source directory. Depending on your development environment, you may need to rename first line **#include "pch.h"** to something like **#include "framework.h"**. Verify that you have included the header files from framework.h and added links to the libraries PropSys.lib and ShlWapi.lib. At this point, you should be able to develop your own code.

5. Python Examples

5.1 Description

The python examples folder contains Python scripts that call the Sciencetech COM objects.

The following scripts are included:

- TestCOM_Chopper calls the SciChopper_API COM object.
- TestCOM_Detector calls the SciDetector_API COM object.
- TestCOM_FilterWheel calls the SciFilterWheel_API COM object.
- TestCOM_Iris calls the SciIris_API COM object.
- TestCOM_LampPower calls the SciLampPower_API COM object.
- TestCOM_Mono calls the SciMono_API COM object.

The general structure for the Python examples is :

1. Import the win32com.client
2. Define an event handler class. In our simple examples, the event handler prints the event data for **OnError** and **OnPropertyChanged** events.
3. Create an instance of the COM object, for example:
 - a. fw = win32com.client.Dispatch("SciencetechCom.SciFilterWheel_API")
4. Add the event handler using (for example) win32com.client.WithEvents(fw, FirstEventHandler)
5. Set the class name, dll path and configuration files.
6. Call COM object properties and methods. The **OpenConfigWindow** method should open the COM object setup window. If the configuration file does not open, verify the configuration file and DLL paths are correct.

Some of the COM objects have methods with output parameters, for example

GetFilterDescription of SciFilterWheel_API. To call this method in Python, use the following:

```
s = ""  
  
s = (f2.GetFilterDescription(2, s))
```

Filter description will be in parameter s[1].

5.2 Requirements

The computer must have Python **32bitts** installed. Install the pyWin32 Python extensions for windows (<https://pypi.org/project/pywin32/>). Ensure that the correct bit level is used (32 bit or 64 bit).

6. Example of automated sequence

6.1 AutoFilter Selection Procedure

1. Monitor the monochromator **OnPropertyChanged** event.
2. Check for the monochromator **IsMoving** property.
3. If **IsMoving** property has changed to true, call the monochromator **GetWavelength** method to retrieve the wavelength in nm.
4. Call the filter wheel **MoveToFilterWavelength** method with this obtained wavelength to set the appropriate filter.

Appendix A: Class Name

This table provides all the available class name per module.
Please select the class name based on your equipment description.

Table 1 : ClassName description

Product	Assembly File Name	ClassName
Monochromator		
Single Monochromator 9055	Mono9055.dll	DirectDriveMonoControl.Mono9055
Double Monochromator 9055	Mono9055.dll	DirectDriveMonoControl.DirectDriveDX
Single Monochromator SDE 9040	SDEMono9040.dll	SDEMono9040.MonoControl
Double Monochromator SDE 9040	DXSDEMono9040.dll	DXSDEMono9040.DXMono
Trinamic Motor Monochromator	Mono9055_StepperMotor.dll	Mono9055_StepperMotor.Mono9055
FilterWheel		
ACE SDC Filter wheel	SciMotorControl.dll	SciMotorControl.FilterWheel
DMXJ Filter Wheel	SciMotorControl.dll	SciMotorControl.DMXJFilterWheel
Trinamic Filter Wheel	StepperMotor_FilterWheel.dll	StepperMotor_FilterWheel.FilterWheel
Chopper		
CH60	SciChopper.dll	SciChopper.Chopper
MC2000 & MC2000B	MCChopper.dll	MCChopper.Chopper
DETECTOR		
USB 1608FS	MCCDetector.dll	MCCDetector.MCCAcquisition
Power Supply		
PSxxx	Sci601LampPower.dll	Sci601LampPower.LampPower601

BK Power supply	BKPowerSupply.dll	BKPowerSupply.PowerSupply
IRIS		
XILab IRIS	XIMCIris.dll	XIMCIris.Iris
Slit		
Thorlabs Slit	TLSlit.dll	TLSlit.Slit